(Visualizing) Plausible Treatment Effect Paths

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Abstract

We study inference on dynamic treatment effects. We propose simple additions to standard visualizations aimed at enhancing their informativeness. First, we recommend reporting joint hypothesis tests and summary measures that are often of direct interest alongside the standard event-study plot. Second, we introduce a data-driven procedure that estimates a smooth approximation to the treatment-effect path within a prespecified, economically motivated class and produces corresponding confidence regions. The resulting intervals can be substantially tighter than conventional uniform bands while maintaining formal coverage guarantees through post-selection inference (Berk et al. [2013]).

JEL Codes: C12, C13

Keywords: dynamic treatment effects, post-selection inference, uniform inference

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1 Introduction

We study settings where researchers aim to understand the dynamic effects of a policy. For example, a macroeconomist may estimate the response of the Consumer Price Index to a monetary shock via local projections, or a microeconomist may estimate the response of labor market outcomes to a change in welfare policies via distributed lag models. In both cases, the researcher would typically summarize estimated effects over time in a dynamic effect (or event-study) plot.

The top panel of Figure 1 depicts an example as it may appear in the literature. The black dots provide point estimates of effects for eight periods preceding the event and 36 periods following the event. As is standard, the figure includes pointwise 95% confidence intervals given by the inner confidence set, indicated by the dashes. We further include uniform 95% sup-t confidence bands as the outer confidence set. While reporting uniform confidence bands is less standard than reporting pointwise intervals, it is increasingly common to include them (e.g., Callaway and Sant'Anna [2021], Jordà [2023]).

We start from the premise that researchers present dynamic effect plots to visualize what treatment paths are plausible using the displayed estimates and uncertainty measures. In this note, we first argue that this type of "visual inference" may be ineffective under this premise. The displayed 95% pointwise intervals do not provide their stated nominal coverage when the object of interest is the dynamic effect path as a whole. In contrast, the 95% sup-t confidence bands do provide valid inference for the entire path with the stated coverage rate. However, in practice these bands are often very wide because they impose no restrictions on the shape of the dynamic effect path and therefore include trajectories that many researchers would regard as *a priori* unreasonable.[1]

To help address these potential shortcomings and provide a more effective visualization, we propose to *augment* standard dynamic effect plots as illustrated in the bottom panel of Figure 1. Our first recommendation is to explicitly report the results of joint hypothesis tests that are often of primary interest. When a dynamic effect plot includes pre-event periods, the goal is typically to assess the presence of pre-trends. Rather than relying on visual inspection of confidence bands, we propose directly testing the joint null that all pre-event coefficients are equal to zero using a standard Wald (or F) test. We include the outcome of this test in the bottom panel of Figure 1 below the x-axis ("Pretrends p-value"). In the displayed

[1] We review various existing confidence regions in more detail in Section 2.

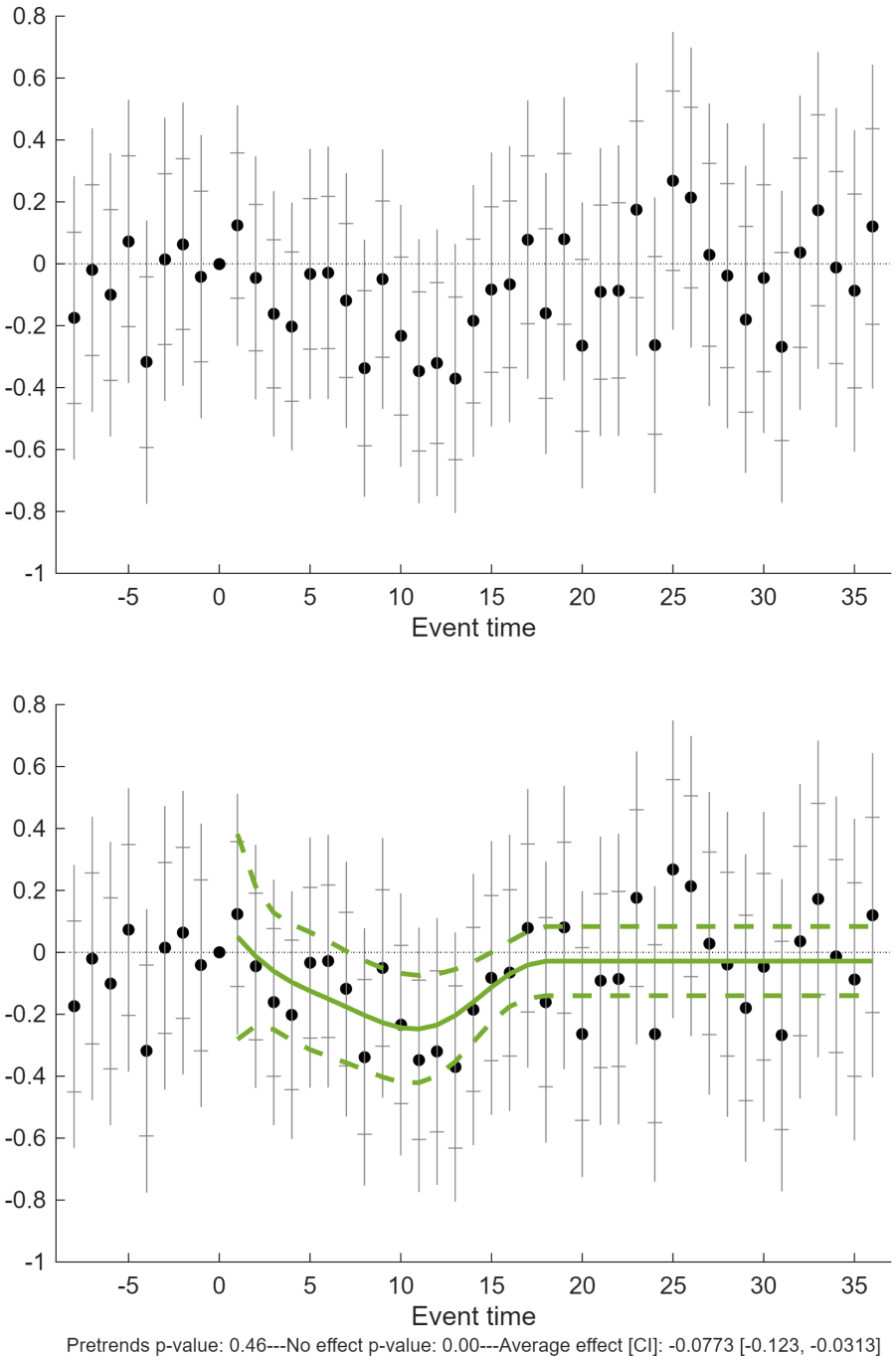


Figure 1: Exemplary dynamic effect plot as it may appear in the literature (top panel), and a version including our proposals (bottom panel). Our main proposal is to include point estimates (the thick solid green line) and corresponding interval estimate (the dashed green lines, which we term *plausible bounds*) of the post-event dynamic effect path obtained by data-driven selection of a smooth model. We further recommend providing results of conventional tests of joint hypotheses that are commonly of interest and an estimate of the average post-event effect.

example, the p-value resulting from this joint hypothesis test is 0.46. Perhaps interestingly, a researcher relying only on pointwise intervals may come to a different conclusion as the pointwise confidence interval at event-time -4 excludes zero. Likewise, we report the p-value from a Wald test of the joint null that all post-event coefficients are equal to zero ("No effect p-value"). Here the conclusions differ between the "visual inference" a reader may do based on the sup-t bands and a simple Wald-test: All uniform confidence intervals after the event include zero, while the joint test strongly rejects the null with a p-value of 0.00. Finally, we report an estimate of the average post-event effect and its corresponding 95% confidence interval as a simple summary of the post-event effects.

Our second, and main, suggestion is to include new visual elements meant to depict approximations to the underlying effect path constrained to exhibit economically and statistically plausible shapes. Intuitively, these elements formalize "visual data snooping": Our proposal approximates looking at a dynamic effect plot and then choosing a specification that "captures the main dynamics" while a) providing a notion of valid inference that explicitly accounts for the data-dependent choice of specification and b) using the full covariance structure of the estimates when choosing the specification.[2]

Specifically, we construct *restricted estimates* and *plausible bounds* by using data-driven regularization to select an approximation to the treatment path from a pre-specified class. We consider a default universe motivated by a preference for smooth dynamics that eventually die out. The restricted estimates are the resulting regularized point estimates of the treatment path. Our plausible bounds are sup-t bands under the selected approximation, adjusted to account for data-dependent selection using Berk et al.'s [2013] post-selection inference (PoSI).

We illustrate the restricted estimates and plausible bounds in Figure 1. The solid green line shows the restricted estimates, and the dashed green lines provide the corresponding plausible bounds. Compared to the standard sup-t intervals, the plausible bounds are relatively narrow and seemingly quite informative about the broad features of the treatment effect path.

The remainder of this note proceeds as follows. We first review existing approaches to describing uncertainty about multiple parameters. We then formally introduce our main

[2]In contrast to the standard pointwise confidence region, the covariance between estimates has important implications for both standard joint tests and our data-driven approximation to the dynamic effect path. We illustrate this in Appendix A.

proposal and conclude with simulation evidence on its performance.

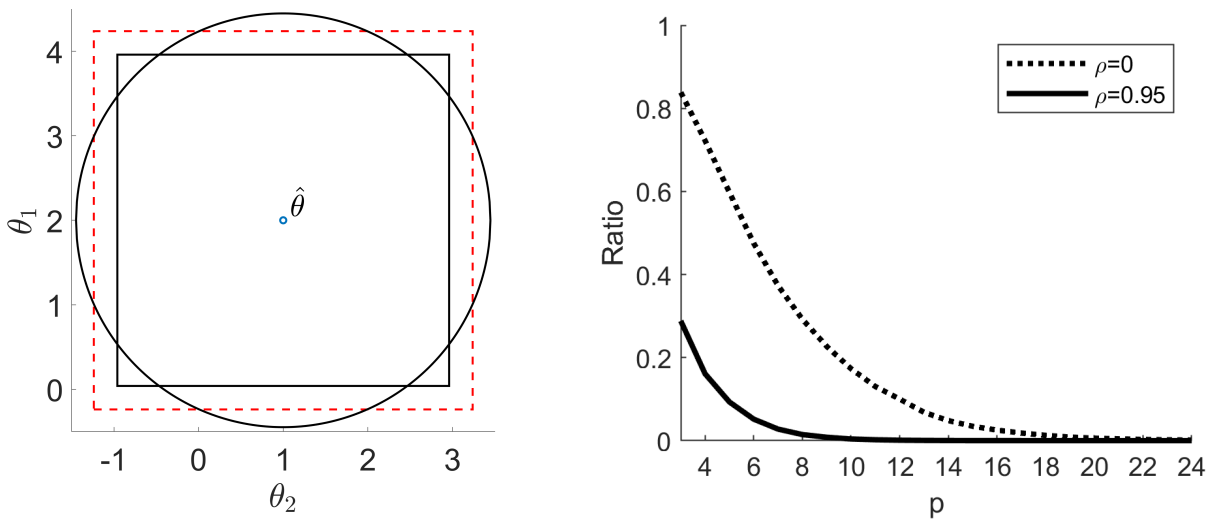
2 Visualizing Uncertainty

We write $\theta = (\beta^{pre'}, \beta')'$ for the $p \times 1$ vector that collects the elements of the dynamic treatment effect path depicted on the dynamic effect plot. Here, $\beta_{pre} = \{\beta_h^{pre}\}_{h=-L}^{-1}$ denotes the displayed pre-event effects, $\beta = \{\beta_h\}_{h=1}^{H}$ collects the dynamic treatment effect path after the event up to the fixed maximum horizon of interest H, and thus $p = H + L$. We assume access to point estimates of the parameters θ_h, denoted by $\hat{\theta}_h$, that correspond to the cumulative effect of the policy at horizon h relative to period 0. Throughout, we assume the vector that collects the estimated dynamic treatment effect path, $\hat{\theta}$, satisfies $\hat{\theta} \sim N(\theta, V_\theta)$ and that we have access to the covariance matrix V_θ. Leading examples to obtain such estimates include distributed lag models, local projections, and event studies.[3]

The standard approach in economics to quantify and visualize the uncertainty associated with an estimate is to construct a confidence region. Figure 2a illustrates common confidence regions for a two dimensional parameter $\theta = (\theta_1, \theta_2)$. The figure uses illustrative values $\hat{\theta} = (2, 1)$ and $V_\theta = I_2$, where I_2 denotes the 2×2 identity matrix.

The most commonly reported confidence intervals are pointwise intervals. $100(1-\alpha)\%$ pointwise intervals for a specific θ_h simply correspond to choosing $(\ell_h(X), u_h(X)) = (\hat{\theta}_h - z_{1-\alpha/2}\sqrt{V_\theta[h,h]}, \hat{\theta}_h + z_{1-\alpha/2}\sqrt{V_\theta[h,h]})$, where $V_\theta[h,h]$ is the variance of $\hat{\theta}_h$ and $z_{1-\alpha/2}$ is the $1-\alpha/2$ quantile of a standard normal distribution. For example, the pointwise 95% confidence intervals in Figure 2a for θ_1 and θ_2 are, respectively, 2 ± 1.96 and 1 ± 1.96. Correspondingly, the black square depicts the Cartesian product of these pointwise confidence intervals for θ_1 and θ_2. Denote the region defined by the black square as CR^{pw}. Treated as a confidence region for (θ_1, θ_2), CR^{pw} $a)$ ignores any information in the off-diagonal entries in the covariance matrix of $\hat{\theta}$ and $b)$ is only valid for testing pre-specified hypotheses involving single coefficients. Thus, it does not achieve correct coverage for the vector $\theta = (\theta_1, \theta_2)$: For a chosen significance level α, it will generally be true that $\mathbb{P}(\theta \in CR^{pw}) = \mathbb{P}(\ell_h(X) < \theta_h < u_h(X) \ \forall \ h) < (1-\alpha)$, such that the black square will generally cover the true parameter θ in less than $100(1-\alpha)\%$ of realizations of the data. For example, if $Cov(\hat{\theta}_1, \hat{\theta}_2) = 0$, the probability that the pointwise confidence region covers the vector θ will

[3] We abstract away from approximation issues, but note that standard asymptotic approximations along with access to consistent asymptotic variance estimators motivate this setup.



(a) Illustration of Pointwise (black rectangle), sup-t (red dashed rectangle), and Wald (black circle) 95% confidence region in two dimensions. $V_\theta = I_2$.

(b) Volume of Wald confidence region relative to sup-t confidence region as a function of $p = \dim(\theta)$. V_θ is a symmetric Toeplitz matrix with entries $v_{ij} = \rho^{|i-j|}$.

Figure 2: Confidence regions.

be $P(\theta \in CR^{pw}) = (1-\alpha)^2$. An immediate implication is that pointwise intervals do not deliver the stated coverage level for inference on the treatment effect path viewed jointly or for many summaries of interest.

One way to construct a uniformly valid confidence region, a region that provides the nominal coverage guarantee for the entire vector (or any subset), is to take the Cartesian product of sup-t confidence intervals (depicted in red in Figure 2a). Denote this region CR^{sup-t}. Sup-t intervals are easy to visualize and construct, differing from pointwise intervals only through the use of a larger critical value. Specifically, $100(1-\alpha)\%$ sup-t intervals are constructed as $\big(\ell_h(X), u_h(X)\big) = \big(\hat{\theta}_h - c_\alpha\sqrt{V_\theta[h,h]}, \hat{\theta}_h + c_\alpha\sqrt{V_\theta[h,h]}\big)$, where c_α is set such that $\mathbb{P}(\ell_h(X) < \theta_h < u_h(X)\ \forall\ h) \geq (1-\alpha)$. For a chosen significance level α, CR^{sup-t} thus achieves valid coverage, since $\mathbb{P}(\theta \in CR^{sup-t}) \geq (1-\alpha)$ by construction. For example, the sup-t 95% confidence intervals in Figure 2a for θ_1 and θ_2 are, respectively, 2 ± 2.24 and 1 ± 2.24. We focus on pointwise and sup-t confidence intervals, since these two are the predominant intervals used in practice (e.g., Callaway and Sant'Anna [2021]; Jordà [2023]).[4]

Finally, the black circle in Figure 2a corresponds to an alternative confidence region for

[4]In practice, we obtain c_α by simulating from multivariate normal distributions with covariance matrix V_θ. See, e.g., Freyberger and Rai [2018] and Olea and Plagborg-Møller [2019] for further discussion about sup-t intervals and related rectangular confidence regions.

θ, namely the Wald confidence region. Denote this region CR^{Wald}. This region simply collects all parameter values (b_1, b_2) that are not rejected by a standard Wald test of the null hypothesis that $(\theta_1, \theta_2) = (b_1, b_2)$ at level α. For a chosen significance level α, this region also achieves valid coverage for the vector θ: $\mathbb{P}(\theta \in CR^{Wald}) = (1 - \alpha)$ by construction. Unlike the usual pointwise confidence region and the sup-t confidence region, the Wald confidence region is hard to visualize for $p > 2$.[5]

While the volumes of the Wald and sup-t regions can be similar for $p = 2$ (cf. Figure 2a), the volume of the sup-t confidence region quickly explodes relative to the volume of the Wald region as p increases. We illustrate this divergence in Figure 2b, which plots the volume of the Wald confidence region relative to the volume of the sup-t confidence region as a function of the dimension p for two exemplary covariance matrices V_θ. We see that the volume of the sup-t region tends to be orders of magnitude larger than the volume of the Wald region for the typical horizon that is depicted in event studies and impulse responses. When V_θ is the identity matrix ($\rho = 0$), the relative volume of the Wald region is less than 10% and around 0.1% of the volume of the sup-t region for $p = 12$ and $p = 24$, respectively. These differences are even more pronounced when the elements of $\hat{\theta}$ are correlated: When V_θ is a symmetric Toeplitz matrix with entries $v_{ij} = 0.95^{|i-j|}$ ($\rho = .95$), this ratio drops to 3.5% and 0.0001% for $p = 12$ and $p = 24$, respectively.

The Wald and sup-t confidence regions are non-nested, implying that neither uniformly dominates the other. However, an immediate consequence of the difference in volume is that, for even moderate p, the overwhelming majority of paths inside the sup-t bands would be rejected by a standard joint hypothesis test. This property seems unappealing to us, and it suggests that sup-t confidence bands may not always provide an informative visualization of which dynamic treatment effect paths are plausible. Figure 1 illustrates this tension. The sup-t region includes treatment paths that imply an overall positive effect (i.e., paths with $\sum_{h=1}^{H} \beta_h > 0$), as well as paths with markedly different shapes. Indeed, $\theta = 0$ lies within the sup-t bands, suggesting that the null hypotheses of "no effect" and "no pre-trends" are both plausible. In contrast, the Wald test of the joint null $\beta_h = 0$ for $h = 1, \ldots, H$ yields a p-value of 0.00. More generally, one might reasonably conclude from the sup-t region alone that the data are largely uninformative about the post-event effects β. Yet, if we uniformly sample treatment paths from the sup-t region in Figure 1, only about one path in 100,000 is not rejected by a joint test (cf. Figure 2b). Consistent with this, the 95% confidence interval

[5]See Appendix A for an illustration of confidence regions under non-zero correlation.

for the average effect is $[-0.12, -0.03]$, suggesting a negative overall effect.

These discrepancies between the easy-to-visualize sup-t region and the results of more standard hypothesis tests lead to our first recommendation for enhancing the informativeness of dynamic effect plots. When there are clear null hypotheses or summaries of interest (e.g., the absence of pre-trends, the absence of post-event effects, or the overall average effect), we advocate reporting the results of the corresponding joint hypothesis test or summary along with the sup-t visualization. The two convey complementary information. The sup-t bands offer a uniformly valid, compact visual summary of the estimated path and its precision, while the additional statistics directly address specific hypotheses or targets of interest. Reporting only summaries omits information about the shape and uncertainty of the full path, while reporting only the sup-t bands provides insufficient information to recover the results of standard procedures targeting specific objects.

There are also many settings in which no single, well-defined null hypothesis is of primary interest. A leading example is the analysis of post-event dynamics, where researchers may not have a specific parametric structure or reference path in mind. In such cases, inference based on visualization of the dynamic effect plot seems particularly compelling. In the next section, we propose an approach designed to improve informativeness in this setting.

3 Plausible Bounds

Our main proposal is to augment the dynamic effect plot with an estimated post-event effect path and corresponding confidence region that covers approximations of the true path that have "reasonable shapes."[6] We refer to these confidence regions as *plausible bounds* and to the corresponding point estimates as *restricted estimates*. We operationalize "reasonable shapes" by pre-specifying a universe of candidate representations for the post-event path. We then use data-dependent selection procedure to choose a good representation for $\hat{\beta}$ from among this set. Intuitively, this approach is related to directly imposing a functional form restriction as is often done in empirical work, for example, by

- specifying a parametric model for β, e.g., imposing a constant treatment effects model

[6] We focus exclusively on post-event effects in this section because there is typically a well-defined null hypothesis of interest for pre-event effects, and we do not expect smoothing of pre-event coefficients to be desirable in most settings. However, pre-event effects could easily be incorporated if desired.

$(\beta_h = \beta_{h'} \ \forall \ h, h')$,

- aggregating the underlying dataset over time (e.g., monthly to quarterly), which effectively restricts β to "step functions,"
- estimating an impulse response function (IRF) via a vector auto regression (VAR), which restricts the IRF to functional forms compatible with the chosen VAR (cf. the discussions in Li et al. [2024] and Montiel Olea et al. [2024]).

One key feature of our approach is that we do not rely on a fixed functional form restriction or on some other implicit device to restrict the dynamic effect path. Rather, we use the data to choose a representation from among a pre-specified set of candidates. We then take this data-dependent choice explicitly into account when constructing confidence bounds. That is, we propose a procedure that is explicit, transparent, and allows us to maintain formal coverage guarantees despite relying on the data to choose an approximation to the dynamic effect path.

Before formally defining our proposal, we introduce the notion of a "surrogate" from the nonparametric statistics literature (cf. Genovese and Wasserman [2008]) and define some notation. A surrogate path β_M is a population object that approximates the true treatment path β, while potentially being simpler in structure. For example, we may define a constant treatment effects surrogate of β as $\beta_s = \arg\min_b \ (\beta - b)'(\beta - b) \ \text{ s.t. } b_h = b_{h'} \ \forall \ h, h'$. If the surrogate model M is fixed *a priori* (and not itself a function of the data), inference for β_M is straightforward, though we stress that any inferential statements in this case will be about β_M and not β.[7] However, problems for inference arise when the surrogate is selected in a data-dependent manner (e.g., Leeb and Pötscher [2005] or Roth [2022]). In our setting the surrogate is explicitly a function of the data (or more precisely, of the unrestricted estimates $\hat{\beta}$), and we thus write $\beta_{M(\hat{\beta})}$ to denote a data-dependent surrogate path. This observation motivates a two-step procedure. In a first step, we use the data to select the surrogate model. In a second step, we then create a uniformly valid confidence region for the selected surrogate path, taking into account that the choice of surrogate is also random (i.e., a function of the data).

[7] Targeting a simple surrogate function is akin to the standard approach in economics of estimating linear models even when the conditional expectation function is not believed to be linear. One can think of the linear model as a "surrogate model" capturing the best linear predictor. Inference will then be about the linear surrogate, and not the "truth."

Given that we select a representation from a pre-specified universe, a key choice is the specific universe we consider. We focus on a universe motivated by two simple pieces of economic intuition about the dynamics of treatment effects:

1. Treatment effects eventually stabilize. That is, after some horizon K, the treatment effect is constant. We treat K as unknown and allow K to be as large as H, thus allowing rich dynamics across the entire depicted horizon.

2. The dynamic treatment path is "smooth," in the sense that abrupt, high-frequency oscillations in the estimated path are viewed as unlikely. We formalize this notion of smoothness using third differences of the treatment path.[8]

In practice, we consider surrogates with varying degree of shrinkage over first and third differences of $\hat{\beta}$ to implement 1. and 2.[9]

Formally, we assume that the estimator of the treatment path $\hat{\beta}$ is jointly normal with $\hat{\beta} \sim N(\beta, V_\beta)$, where $V_\beta = \sigma^2 V$ and V is positive-definite and normalized so that $\frac{1}{H}\sum_{h=1}^{H} V(h,h) = 1$. Taking $\hat{\beta}$ as input, we define a class of restricted estimators:

$$\begin{aligned}\tilde{\beta}(\lambda_1, \lambda_2, K) &= \arg\min_b \; Q(b, \lambda_1, \lambda_2, K) \\ &= \arg\min_b \; \underbrace{(\hat{\beta} - b)' V^{-1} (\hat{\beta} - b)}_{\text{distance from } \hat{\beta}} + \underbrace{\lambda_1 b' D_1' W_1(K) D_1 b}_{\substack{\text{penalty on first differences} \\ \text{after horizon K}}} + \underbrace{\lambda_2 b' D_3' W_3 D_3 b,}_{\substack{\text{penalty on} \\ \text{third differences}}} \end{aligned} \tag{1}$$

where

- D_1 and D_3 are $(H-1 \times H)$ and $(H-3 \times H)$ matrices such that $D_1 b$ and $D_3 b$ are respectively the vectors of the first and third differences of ordered vector b,

- $(\lambda_1,\ \lambda_2,\ K)$ are tuning parameters,

- $W_1(K)$ and W_3 are weighting matrices, where $W_1(K)$ only places weight on first differences for horizon $h \geq K$ and beyond (see Appendix B for further details).

[8]For a sequence $\{\beta_h\}$, the first difference at horizon h is $\Delta\beta_h = \beta_h - \beta_{h-1}$. The third difference is defined recursively as $\Delta^3\beta_h = \Delta(\Delta(\Delta\beta_h)) = \beta_h - 3\beta_{h-1} + 3\beta_{h-2} - \beta_{h-3}$.

[9]Penalizing higher-order differences can be viewed as a discrete analogue of higher-order derivative penalties in smoothing splines, with closely related formulations appearing in higher-order trend filtering; see, e.g., Wahba [1990], Green and Silverman [1994], Kim et al. [2009].

The first term in (1) measures the distance between the restricted path b and the unrestricted estimates $\hat{\beta}$, while the remaining terms impose smoothness and stabilization as discussed above.

Solving (1) provides a closed form solution for $\tilde{\beta}(\lambda_1, \lambda_2, K) := \tilde{\beta}(M)$:

$$\begin{aligned}\tilde{\beta}(M) &= \left(V^{-1} + \lambda_1 D_1' W_1(K) D_1 + \lambda_2 D_3' W_3 D_3\right)^{-1} V^{-1}\hat{\beta} \\ &= P(M)\hat{\beta}.\end{aligned}$$

For a fixed choice of $M = (\lambda_1, \lambda_2, K)$, it immediately follows that

$$\tilde{\beta}(M) - P(M)\beta \sim N(0, V_M), \tag{2}$$

where $V_M = P(M)V_\beta P(M)'$. Here, $P(M)\beta = \beta_M$ defines a particular surrogate path for β. Intuitively, $P(M)\beta$ corresponds to a "projection" of the true treatment path β onto a restricted class of paths implied by M.

Given (2), constructing a confidence region for the components $\{\beta_{M,h}\}_{h=1}^{H}$ would be straightforward if the tuning parameters were fixed *ex ante*. In practice, however, it is unclear how to choose values for λ_1, λ_2, and K without reference to the data. We therefore use a model selection procedure to choose these tuning parameters (or, equivalently, to select the surrogate model M). Specifically, we use the estimated $\hat{\beta}$ and an object akin to an information criterion to select the surrogate M. First, note that we can construct the "residuals" $\hat{\beta} - \tilde{\beta}(M) = \hat{\beta} - P(M)\hat{\beta} = (I - P(M))\hat{\beta}$ for any candidate M. We use this residual formulation to define an analog of model degrees of freedom given by $\text{df}(M) = \text{trace}(P(M))$. We then select a surrogate model by minimizing a BIC analog over the $\mathcal{M}$, where $\mathcal{M}$ denotes the universe of considered values for $M = (\lambda_1, \lambda_2, K)$:

$$\hat{M} = \underset{M \in \mathcal{M}}{\arg\min}\ (\hat{\beta} - \tilde{\beta}(M))' V_\beta^{-1} (\hat{\beta} - \tilde{\beta}(M)) + \log(H)\text{df}(M).$$

We discipline the selection procedure by pre-specifying $\mathcal{M}$, the universe of models considered. In our implementation, $\mathcal{M}$ includes surrogate models corresponding to paths of the form $P(M)\beta = \beta_M$ using a grid over $(\lambda_1, \lambda_2, K)$. We further extend our model universe by also including surrogate models corresponding to a constant, linear, quadratic, and cubic treatment effect path (with one, two, three, and four degrees of freedom, respectively), as well as an unrestricted model corresponding to the unrestricted estimates $\hat{\beta}$ (with H degrees of freedom). We discuss our implementation in more detail in Appendix B and visualize the

part of our model universe corresponding to the grid over $(\lambda_1, \lambda_2, K)$ for exemplary treatment paths β in Appendix Figure 8. Importantly, $\mathcal{M}$ is fixed *ex ante* and does not depend on $\hat{\beta}$ or σ.

Remark 1. *One could use other model universes and shrinkage methods; examples include Barnichon and Matthes [2018] and Barnichon and Brownlees [2019]. We have chosen a class that we believe provides a reasonable representation of beliefs in many applications. Our approach with this model class is also computationally convenient, which allows us to consider a large universe of candidate representations. Likewise, alternative criteria that trade off fit and complexity (e.g., URE or AIC) could be used to select the surrogate model M in place of our BIC-type criterion. We adopt a BIC-type criterion as a default because it tends to favor more parsimonious representations than other common alternatives. We view this feature as particularly appropriate in a visualization context where the goal is to summarize dynamic patterns. Importantly, our inferential procedure does not rely on the optimality of either choice, and users with substantive reasons to prefer alternative universes or selection criteria can readily substitute them.*

Given the selected surrogate $\hat{M}$, we define the *restricted estimates* as $\tilde{\beta}(\hat{M})$. However, we cannot directly apply (2) to obtain a valid confidence region for the population surrogate path $\beta_{\hat{M}} = P(\hat{M})\beta$ because $\hat{M}$ was selected by looking at the data, $\hat{\beta}$. To account for this data-dependent (and therefore random) selection, we proceed in a second step by using valid post-selection inference (PoSI; Berk et al. [2013]) to construct a uniformly valid confidence region for $\beta_{\hat{M}}$.

The resulting confidence intervals, which we refer to as *plausible bounds*, are rectangular regions of the form $CR^{PoSI} = \{\ell_h(X), u_h(X)\}_{h=1}^{H}$ for $[\ell_h(X), u_h(X)] = [\tilde{\beta}(\hat{M})_h \pm C^{\alpha} V_{\hat{M}}^{1/2}(h,h)]$. Here, $\tilde{\beta}(\hat{M})_h$ denotes the restricted estimate of the effect at horizon h and $V_{\hat{M}}^{1/2}(h,h)$ is the square root of the hth diagonal entry of $V_{\hat{M}}$. To ensure uniform validity, we set C^{α} equal to the "PoSI constant" of Berk et al. [2013], defined as the smallest value satisfying

$$\mathbb{P}\left(\max_{M \in \mathcal{M}} \max_{h} |t_{h \cdot M}| \leq C^{\alpha}\right) \geq (1-\alpha),$$

where $t_{h \cdot M} = V_M^{-1/2}(h,h)\xi_h$, and ξ_h is the hth element of multivariate normal vector ξ with mean $\mathbf{0}_H$ and variance V_M. The inner maximum over h corresponds to a standard sup-t adjustment across event-time horizons for a fixed model, while the outer maximum over M ensures validity uniformly over all surrogate models in $\mathcal{M}$, regardless of which model is

ultimately selected. Importantly, C^α depends only on the pre-specified model universe $\mathcal{M}$ and not on the particular model selection procedure. In practice, we obtain C^α by simulation.

The following proposition is a direct application of Berk et al. [2013].

Proposition 1. *For any treatment path β, we obtain valid coverage for its surrogate $\beta_{\hat{M}}$:*

$$\mathbb{P}[\beta_{\hat{M}} \in CR^{POSI}] \geq 1 - \alpha. \tag{3}$$

Proof. This follows immediately from the guarantees in Berk et al. [2013]:

$$\mathbb{P}(\beta_M \in CR^{POSI} | \hat{M} = M) \geq 1 - \alpha.$$

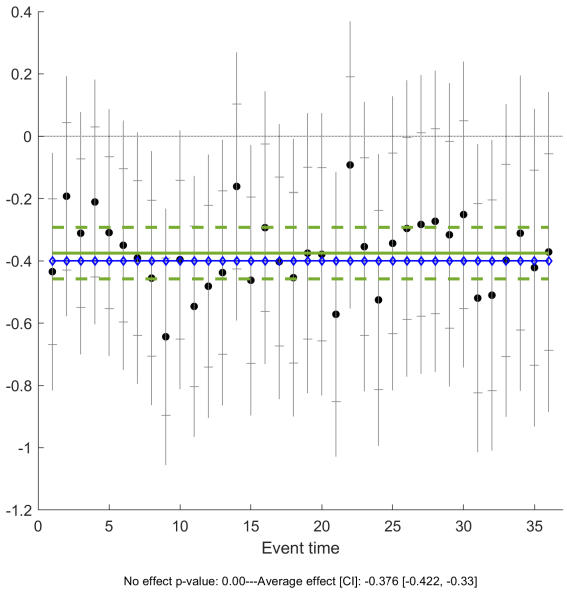
□

Proposition 1 guarantees that the plausible bounds cover the selected surrogate path in at least $(1 - \alpha)\%$ of sample realizations.

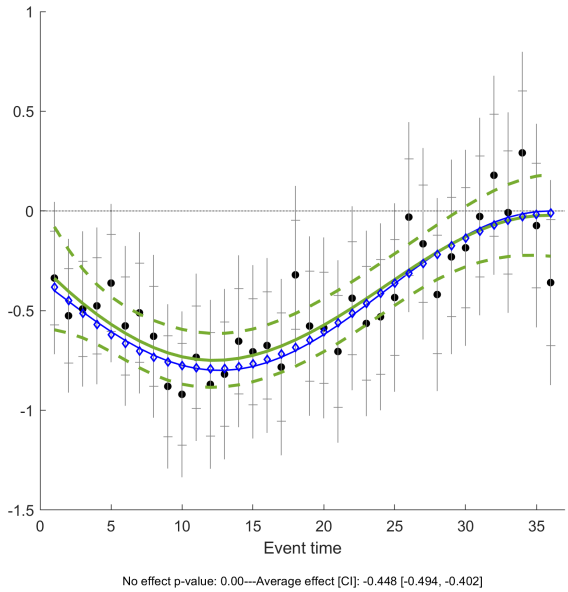
Remark 2. *An immediate consequence of Proposition 1 is that $\mathbb{P}(\beta \in CR^{POSI}) \geq 1 - \alpha$ whenever $\mathbb{P}(\beta_{\hat{M}} = \beta_M = \beta) = 1$. That is, in cases where model selection is effectively non-random and the selected surrogate path coincides with β, the plausible bounds provide coverage for β itself. Given the form of our BIC type objective for selecting $\hat{M}$ and that the unrestricted estimates are always included in our default model universe, one could provide sufficient conditions for $\mathbb{P}(\beta_{\hat{M}} = \beta_M = \beta) = 1$ along sequences of models where $\sigma^2 \to 0$ and surrogate paths are well-separated.*

Remark 3. *Throughout, we work with the unrestricted estimates $\hat{\beta}$ as our goal is to augment existing visualizations that typically start from this point. An alternative would be to estimate the restricted models directly on the data. In settings where $\hat{\beta} \sim N(\beta, V_\beta)$ provides a good approximation, we expect such an approach to yield qualitatively similar results. It may be interesting to explore directly estimating restricted models in settings where this approximation is questionable.*

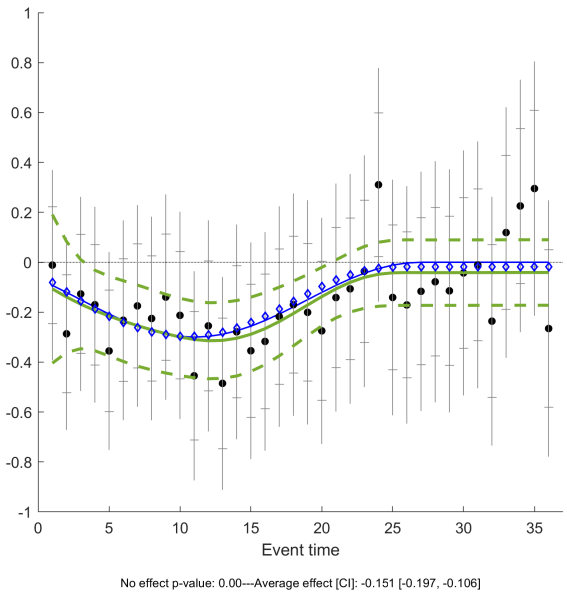
We illustrate our plausible bounds for a single data realization under four hypothetical treatment paths in Figure 3 (see Appendix Figure 4 for additional simulated scenarios, and Appendix F for two exemplary data applications). Relative to Figure 1, we include a solid blue line representing the true treatment effect path and blue diamonds representing the selected population surrogate path. Of course, both objects are unobservable in practice and are included here solely for illustrative purposes.



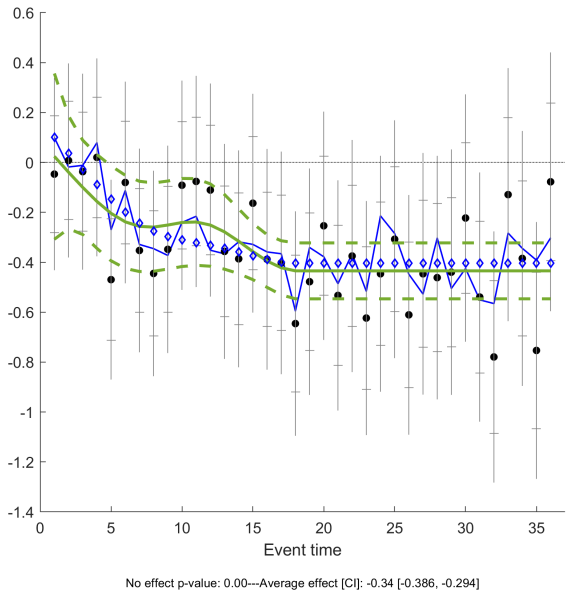
(a) treatment path constant



(b) treatment path hump-shaped, never flat



(c) treatment path hump-shaped, eventually flat



(d) treatment path wiggly

Figure 3: Exemplary treatment effect plots including our proposals. The dashed green lines provide the plausible bounds, and the thick solid green line provides the corresponding restricted estimates. We further include the true treatment path (thin blue line) and its surrogate (dotted blue line).

Looking at the restricted estimates and plausible bounds in each panel paints a starkly different picture compared to the sup-t intervals. In all cases, the plausible bounds are relatively narrow and seemingly quite informative about the broad features of the treatment effect paths. We reiterate that the inferential target of the plausible bounds in the population is *not* the true treatment path. Instead, they provide uniform coverage of the surrogate. In Figures 3a, 3b, and 3c, this surrogate is indistinguishable from the true treatment path. In contrast, the surrogate in Figure 3d clearly differs from the truth but captures what appear to be salient features about the broad trends of the effect dynamics.

An alternative interpretation of our plausible bounds is that they represent a compromise between inference under no restrictions — which may yield very wide confidence intervals — and inference based on a single, prespecified model that does not adapt to the data and may therefore be overly restrictive. We take the visual evidence in Figure 3 (and Appendix Figure 4), together with the simulation results presented in the next section, as evidence that this compromise performs well in practice. We thus argue that adding both sup-t bands and our plausible bounds to a dynamic effects plot provides a more complete picture of the conclusions one might draw about dynamic effect paths. As with any robustness exercise, however, when sup-t bands and plausible bounds point to opposing conclusions, results should be interpreted with caution.

Related Literature. We obtain our plausible bounds and restricted estimates by considering a finite set of candidate representations for β. This approach is thus closely related to work that considers parametric models and approximations to β (e.g., Almon [1965], Barnichon and Matthes [2018], Barnichon and Brownlees [2019]). Our restricted estimates are akin to point estimates that could be obtained by taking an empirical Bayes approach within the framework of Shiller [1973], which considers a closely related model universe and takes a fully Bayesian approach. See also the SmIRF estimator of Plagborg-Møller [2016] for a related approach that includes confidence sets with guaranteed frequentist coverage. In contrast to these papers, our inferential target is the population value of the selected surrogate path; see, for example, Genovese and Wasserman [2008] for a general discussion of inference on surrogates.

Alternative approaches to quantifying and visualizing uncertainty about treatment effect paths include Sims and Zha [1999], who argue that conventional pointwise bands should be supplemented with measures of shape uncertainty. Jordà [2009] suggests a method to construct simultaneous confidence regions for impulse responses given propagation trajectories.

Freyberger and Reeves [2018] propose a uniformly valid inference method for an unknown function or parameter vector satisfying certain shape restrictions. Inoue and Kilian [2016] suggests a "shotgun plot" which depicts a random sample of B impulse responses contained in the joint Wald confidence set to visualize plausible treatment effect paths. We believe our proposal augmenting standard dynamic effect plots with simple additional elements provides a useful complement to this existing literature.

4 Numerical Results

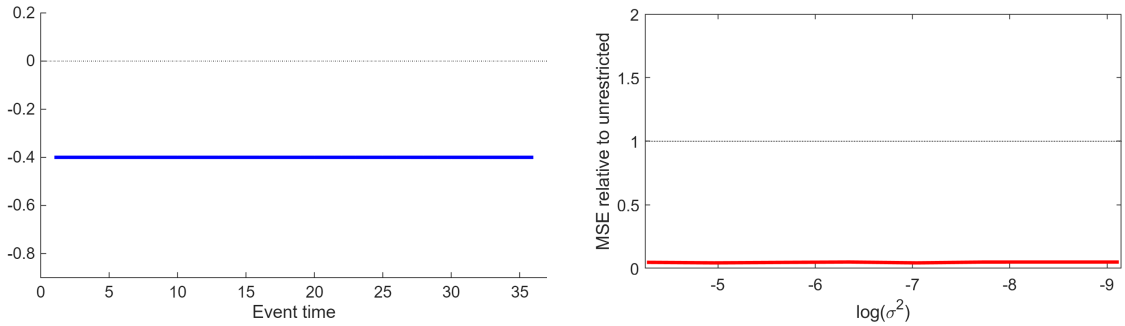
In this section, we illustrate the properties of the restricted estimates and the plausible bounds in simulation experiments with treatment paths generated to resemble treatment path dynamics that practitioners may encounter. We illustrate these treatment paths in the left panels of Figure 4. We consider a constant treatment effect path (cf. Figure 4a); a smooth treatment path that varies over the entire displayed horizon (cf. Figure 4b); a hump-shaped treatment path with dynamics that die out after 27 periods (cf. Figure 4c); and a "wiggly" treatment path (cf. Figure 4d). We describe the exact DGP for each of the four panels in more detail in Appendix Table 1. The object of interest is the treatment path over a 36-month horizon. We have access to jointly normal estimates $\{\hat{\beta}_h\}_{h=1}^{36}$. For each of these four treatment paths, we then draw 1,000 realizations of $\hat{\beta} \sim N(\beta, V_\beta)$.[10]

We compare the point estimation properties of the unrestricted estimates $\hat{\beta}$ with those of our restricted estimates $\tilde{\beta}(\hat{M})$ in the right panels of Figure 4. These panels depict the ratio of mean squared errors, $MSE_{\tilde{\beta}(\hat{M})}/MSE_{\hat{\beta}}$, as a function of σ^2, which scales the covariance matrix of the estimates, V_β (see Appendix C for more detail).

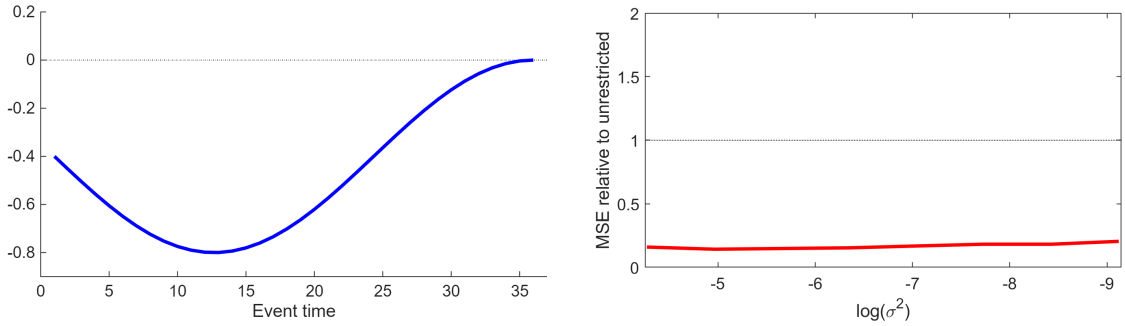
The largest value of σ^2 in the right panels of Figure 4 (corresponding to the leftmost point) represents relatively noisy estimates.[11] Overall, we find that the restricted estimator performs relatively well when the target is the true treatment path β. In the three designs with a smooth treatment path (Figures 4a-4c), the MSE of the restricted estimator is a full order of magnitude lower compared to the unrestricted estimator. The true path is not smooth in

[10] In the figures that follow, V_β is diagonal with its entries specified in Appendix C. Additional results, including those from alternative designs, are reported in Appendix D. We also repeat our exercise with positively correlated estimates in Appendix E.

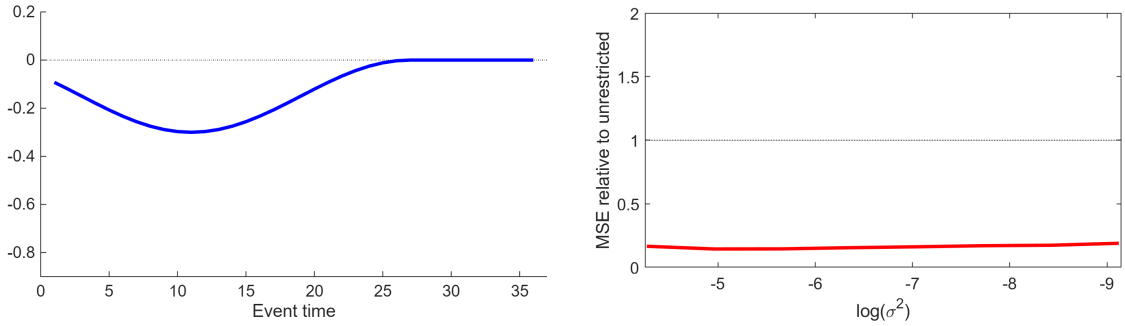
[11] Figure 3 was created from a single realization of the leftmost point in Figure 4. To give the reader a sense of the scale of the x-axes, Appendix Figure 7 also illustrates a single realization of the rightmost point in Figure 4.



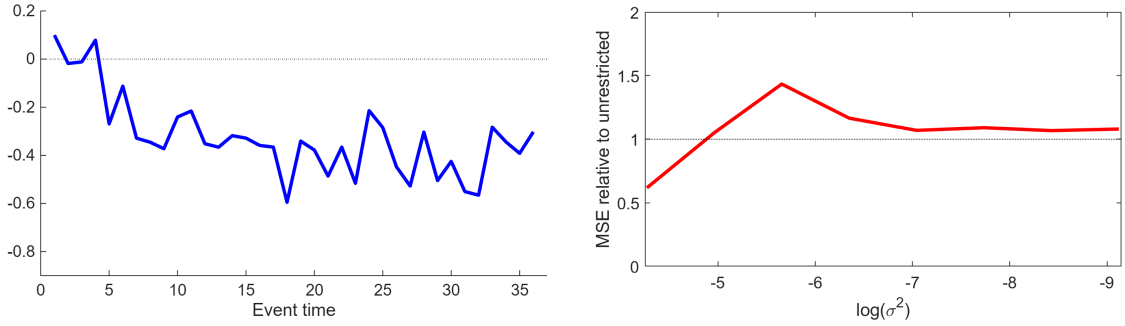
(a) Constant Treatment Effect



(b) Hump-shaped, never flat



(c) Hump-shaped, eventually flat



(d) Wiggly

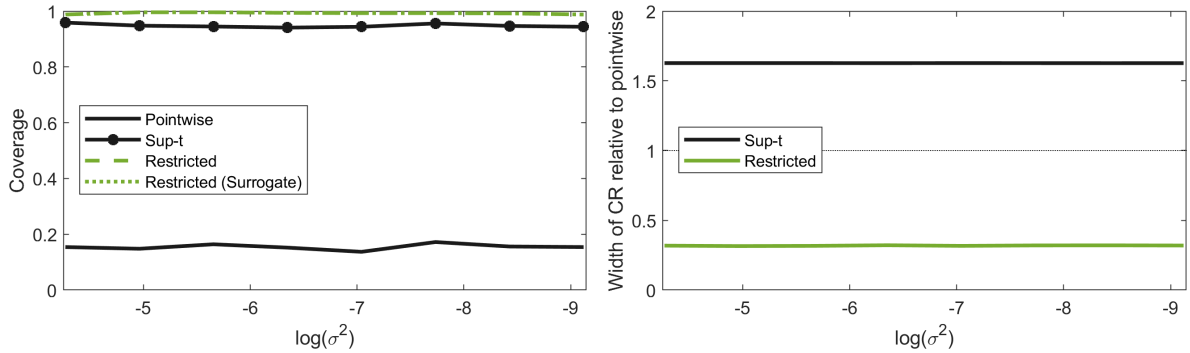
Figure 4: Left panels: Four exemplary treatment effect paths β. Right panels: Ratio of mean squared errors between restricted and unrestricted estimates, $\frac{MSE(\tilde{\beta}(\hat{M}))}{MSE(\hat{\beta})}$, as a function of the amount of noise σ^2 in the initial estimates $\hat{\beta}$.

Figure 4d. In this case, the performance of the restricted estimator in estimating the true path varies with the noise level. Specifically, it outperforms the unrestricted estimator at high noise levels, underperforms at intermediate levels, and performs similarly when noise is low, consistent with changing bias–variance tradeoffs for estimating the true path.

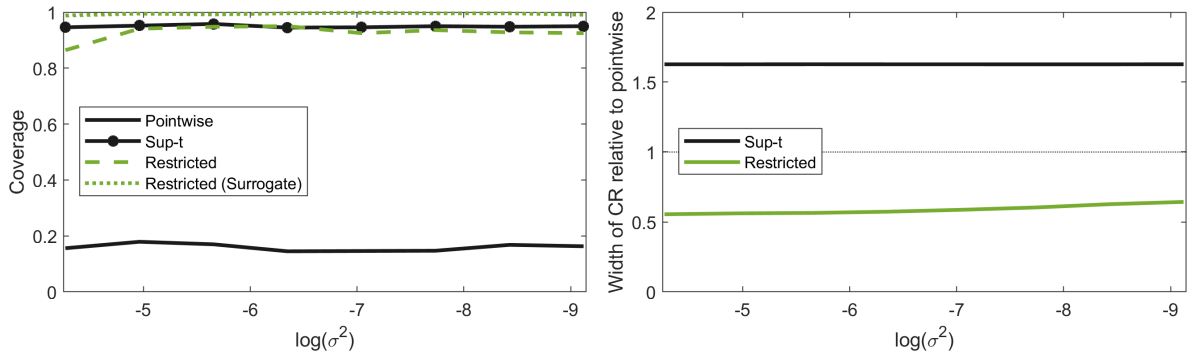
Figure 5 summarizes inference results, where we set $\alpha = 0.05$. The left panels of Figures 5a-5d display coverage results. The pointwise, sup-t, and restricted coverage rates, depicted by the three solid lines, represent the empirical analogue of uniform coverage for β: the fraction of simulations in which the true treatment path β falls entirely inside the pointwise confidence intervals (black), sup-t intervals (black solid line with dots), and plausible bounds (green dashed), respectively.

Across all four DGPs, the pointwise confidence intervals cover the true treatment path in only 15-20% of simulations. In contrast, the sup-t intervals achieve nominal coverage across all DGPs. The plausible bounds are not constructed to provide uniform coverage for the true treatment path. It is thus not surprising that these bounds do not cover the entire treatment path in 95% of simulations across all DGPs. However, their coverage appears to converge towards 95% as the amount of noise in the initial estimates decreases (cf. Remark 2). Further, the plausible bounds perform substantially better than the pointwise confidence intervals in covering the true treatment path when the true path is smooth, achieving near nominal coverage in most cases. Proposition 1 guarantees that the plausible bounds cover the selected surrogate to the truth in at least 95% of realizations. In Figures 5a-5d, the dotted green line illustrates the plausible bounds' coverage of the surrogate. We see that this coverage is above 95% for all levels of σ^2 across all DGPs.

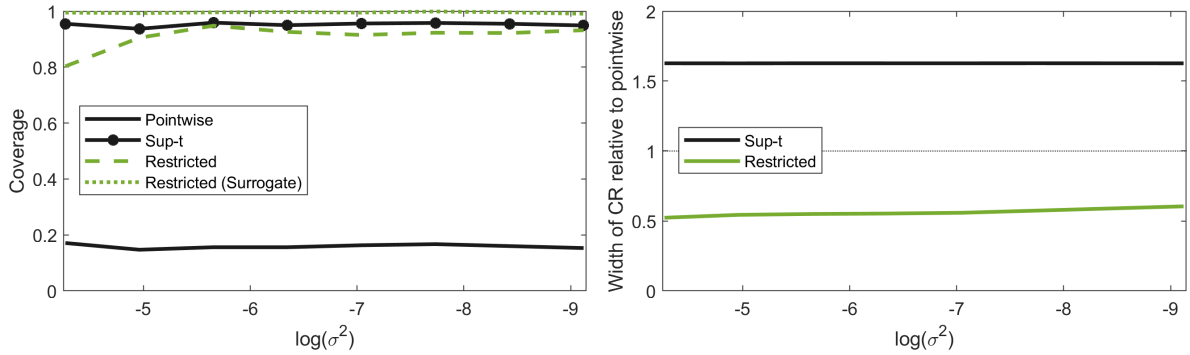
To achieve higher coverage of the true path, the sup-t bands are generally wider than the pointwise bands (cf. Figure 3). We illustrate this tradeoff in the right panels of Figure 5, which depict the average width of the sup-t bands and the plausible bounds relative to the pointwise intervals. Across most scenarios, the plausible bounds are narrower than pointwise and sup-t bands, while offering improved coverage relative to pointwise intervals and often performing comparably to sup-t bands. For example, in the constant treatment effect case, plausible bounds are less than half as wide as pointwise bands and less than a quarter as wide as sup-t bands, yet still achieve close to 95% coverage. The plausible bounds are only wider than the sup-t bands when both (i) the treatment path is wiggly and (ii) there is very little noise. Under these conditions, our model selection mechanism results in relatively little smoothing. The plausible bounds are then slightly more conservative than the sup-t bands,



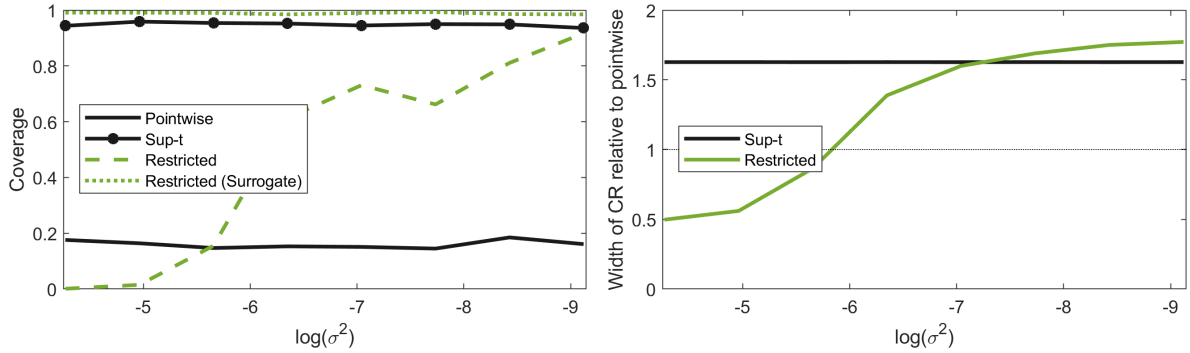
(a) Constant Treatment Effect



(b) Hump-shaped, never flat



(c) Hump-shaped, eventually flat



(d) Wiggly

Figure 5: Left panels: Coverage of various confidence regions as a function of the amount of noise σ^2 in the initial estimates $\hat{\beta}$. Right panels: Width of sup-t and plausible bounds relative to the width of pointwise intervals.

which follows immediately from the results in Berk et al. [2013].[12]

5 Conclusion

We propose simple additions to dynamic effect plots to improve their informativeness. First, we recommend reporting numeric summaries, such as results of standard joint tests of focal hypotheses and average effects, alongside the standard plot. Second, we introduce a new visualization for dynamic treatment effect paths that combines the notion of a *surrogate* from the nonparametric statistics literature (cf. Genovese and Wasserman [2008]) with post-selection inference (Berk et al. [2013]). Our data-driven smoothing delivers a smooth estimate with potentially improved point estimation properties and confidence regions covering this surrogate that can be substantially tighter than conventional pointwise or uniform bands.

It may be interesting to further study inference that explicitly targets data-dependent surrogates in economically relevant settings. While we focused on treatment path estimation, the general approach of shrinking unrestricted models toward simpler representations applies more broadly. For example, one could consider shrinking estimates towards theoretical restrictions implied by economic theory as in Fessler and Kasy [2019]. It would also be interesting to explore related ideas in more structural settings, such as selecting among or regularizing moment conditions in GMM-type frameworks and focusing inference on the selected or regularized objects.

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[12]We illustrate the model selection behavior of our procedure further in Appendix Figures 9-11. Surrogate paths β_M with significantly reduced degrees of freedom are often selected. This effective dimension reduction explains both the improvement in point estimation properties of $\tilde{\beta}(\hat{M})$ relative to the unrestricted estimates $\hat{\beta}$ and the reduction in interval width.

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